Learning from Big Data: Assignment 1
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Group number:

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Introduction

This file provides a LaTeX template for Assignment 1 of the Learning from Big Data course. It was prepared to save you time so that you can focus on the theory and technical parts of the methods seen in class.

You have received the dataset of reviews, the three dictionaries with the training set of words for each topic, a list of stopwords, and a validation dataset containing sentences classified by a panel of human judges. Run your analysis in R to produce your predictions, likelihoods, tables, and figures, then report and interpret them here. This template does not reproduce any R code – it is where you write up and discuss the results of the analysis you ran separately.

Sections marked **QUESTION** are the parts you are asked to answer. Text shown as *[YOUR ANSWER HERE: ...]* marks where you should insert your own writing, tables, or figures.

This report has the following structure:

1. **General Guidelines**
2. **Research Question**
3. **Data aggregation and formatting**
4. **Supervised Learning – The Naive Bayes classifier (NBC)**
5. **Supervised Learning – Inspect the NBC performance**
6. **Unsupervised Learning – Predict using LDA**
7. **Unsupervised Learning – Predict using Word2Vec**
8. **Analysis – answering the research question**
9. **OPTIONAL – run and interpret the AFINN lexicon for sentiment**
10. **Appendix**

Before going further: make sure you have run your analysis in R (Tutorial 0 provides installation instructions) and have the outputs (predictions, likelihoods, tables, figures) ready to paste into this report.

1 General Guidelines

Page length. The template has 4 pages. You are allowed to add 7 to 10 pages, not including the appendix. There is a limit of pages, but you have the possibility of using appendices, which are not limited in number of pages. Use your pages wisely. For example, having a table with 2 rows and 3 columns that uses 50% (or even 25%) of a page is not an efficient use of space.

2 Submission

Make sure that the files you submit on canvas contain the following:

1. The pdf of your report named **Assignment_1_GROUPNUMBER.pdf**
2. Your code, that can be run as-is, which prints all tables/figures you use in your pdf report.
3. If you use any data not on the github, upload it as well.
4. The *Reviews_Raw* file containing the calculated posterior content and sentiment likelihoods.

3 Research Question

QUESTION I. Present here the main research question you are going to answer with your text analysis. You are free to choose the problem and change it until the last minute before handing in your report. However, your question should not be so simple that it does not require text analysis. For example, if your question can be answered by reading two reviews, you do not need text analysis; all you need is 10 seconds to read two reviews. Your question should not be so difficult that you cannot answer in your report. Your question needs to be answered in these pages.

[YOUR ANSWER HERE:]

4 Data aggregation and formatting

QUESTION II. Decide on how to aggregate or structure your data. The data you received is at the review level (i.e., each row is a review). However, the variables in the data are very rich and allow you to use your creativity when designing your research question. For example, there are timestamps, which allow you to aggregate the data at the daily level or even hourly level. There is information on reviewers, which allows you to inspect patterns of rating by reviewer. There is information on studios, and more. Please explicitly indicate how you structured your dataset, and what your motivation is for doing so. Even if you are using the data at the review level, indicate how and why that is needed for your research question.

[YOUR ANSWER HERE:]

5 Supervised learning – the Naive Bayes classifier

5.1 Likelihoods

QUESTION III. Create the content likelihoods based on the three lists of words provided (storyline, acting, visual). Be explicit on the decisions you took in the process, and why you made those decisions (e.g., which smoothing approach you used).

[YOUR ANSWER HERE:]

QUESTION IV. Locate a list of sentiment words that fits your research question. For example, you may want to look just at positive and negative sentiment (hence two dimensions), or you may want to look at other sentiment dimensions, such as specific emotions (excitement, fear, etc.). TIP: Google will go a long way finding these, but do check if there is a paper you can cite that uses your list.

[YOUR ANSWER HERE:]

6 Supervised Learning – Inspect the NBC performance

6.1 Compute confusion matrix, precision and recall

QUESTION V. Compare the performance of your NBC implementation (for content) against the judges' ground truth by building the confusion matrix and computing the precision and accuracy scores. Do not forget to interpret your findings.

[YOUR ANSWER HERE:]

Table 1: Confusion matrix – content classifier vs. judges' ground truth (example layout).

		Predicted		
		Storyline	Acting	Sound/Visual
Actual	Storyline	*[YOUR ANSWER HERE:]*	*[YOUR ANSWER HERE:]*	*[YOUR ANSWER HERE:]*
	Acting	*[YOUR ANSWER HERE:]*	*[YOUR ANSWER HERE:]*	*[YOUR ANSWER HERE:]*
	Sound/Visual	*[YOUR ANSWER HERE:]*	*[YOUR ANSWER HERE:]*	*[YOUR ANSWER HERE:]*

7 Unsupervised Learning: Predict using LDA

QUESTION VI. Using LDA, predict a variable in your dataset (in the case of movie reviews, use movie box office). Tip: you can pass a list of reviews to the LDA package in order to get the posterior probability that the reviews are about each topic. If you pass them all in a single document, you will not get review-specific vectors.

[YOUR ANSWER HERE:]

8 Unsupervised Learning: Predict using Word2Vec

QUESTION VII. Using Word2Vec, predict a variable in your dataset (in the case of movie reviews, use movie box office).
Tip 1: you can reduce the dimensionality of the Word2Vec output with PCA/Factor analysis. This will save you computing time.
Tip 2: Word2Vec gives you word vectors. You can then compute the average of these word vectors for all words in a review. This gives you a vector describing the content of a review, which you can use as your constructed variable(s).

[YOUR ANSWER HERE:]

9 Analysis – Use the constructed variables to answer your research question

QUESTION VIII. Now that you have constructed your NLP variables for sentiment and content using supervised and unsupervised methods, use them to answer your original research question.

[YOUR ANSWER HERE:]

10 OPTIONAL: run and interpret sentiment with the supervised learning AFINN lexicon

QUESTION IX (optional). Using the AFINN code you received in the lecture, compute the sentiment using the syuzhet package/lexicon. Compare the performance of your NBC implementation (for sentiment) assuming that the AFINN classification is the ground truth, then build the confusion matrix and compute precision and recall. Note that we are now interested in understanding how much the two classifications differ and how, but we are not implying that AFINN is error-free, far from it. We are interested in uncovering sources of systematic differences that can be attributed to the algorithms or lexicons. Do interpret your findings.

[YOUR ANSWER HERE:]

Table 2: Confusion matrix – NBC sentiment vs. AFINN (example layout).

		Predicted (NBC)	
		Positive	Negative
AFINN	Positive	*[YOUR ANSWER HERE:]*	*[YOUR ANSWER HERE:]*
	Negative	*[YOUR ANSWER HERE:]*	*[YOUR ANSWER HERE:]*

A Appendix

Use this section (unlimited pages) for supplementary material: extra tables/figures, robustness checks, additional prediction samples, or a link to your GitHub repository in lieu of uploading your code separately.

[YOUR ANSWER HERE:]